

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C04: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 4

Total: Marks:40

Annexure A

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Code of Conduct:

I NARAPPAGARI SRAVYA(192472192)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:sravya

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

ANNEXURE A

INDUSTRY PROBLEM TASK

ANNEXURE A – INDUSTRY PROBLEM TASK

Industry Scenario

A healthcare company wants to predict diabetes using patient features such as age, blood pressure, cholesterol, glucose, BMI, and family history. It also wants to use Reinforcement Learning to recommend personalised actions such as Diet, Exercise, Medication, and Monitor.

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning

Inductive learning learns general patterns from labelled patient data and uses them to predict unseen patients.

Target variable: Diabetes

- 0 – Non-diabetic
- 1 – Diabetic

Important features:

1. **Glucose** – Strong indicator of diabetes.
2. **BMI** – Higher BMI is associated with increased diabetes risk.
3. **Age** – Diabetes risk generally increases with age.
4. **Blood Pressure** and **family history** can also provide useful information.

(b) Handling Class Imbalance

If non-diabetic cases are much higher than diabetic cases, the dataset is imbalanced.

SMOTE is a suitable approach because it creates synthetic samples for the minority diabetic class without removing useful majority-class data.

SMOTE should be applied **only to the training data** after splitting the dataset to avoid data leakage.

Recall should be given importance because missing a diabetic patient (**False Negative**) can be clinically serious.

(c) 70:30 Split and Preprocessing

The dataset is divided into:

- **70% Training**
- **30% Testing**

A stratified split should be used to maintain the same diabetic/non-diabetic proportion.

Preprocessing includes:

- Handling missing values.
- Encoding categorical features such as family history.
- Standardising numerical features.
- Applying SMOTE to the training set.

Standardisation is especially useful for Logistic Regression, while Decision Trees generally do not require it.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree

A Decision Tree classifies patients by repeatedly selecting features that best separate diabetic and non-diabetic cases.

Two common criteria are:

Gini Index:

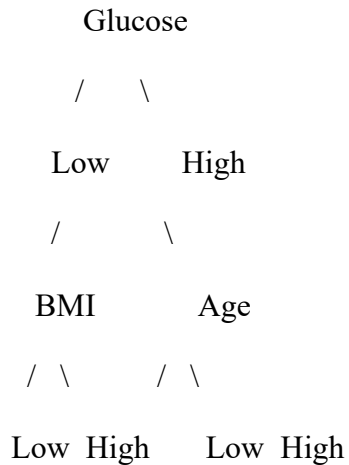
$$\begin{aligned} &[\\ \text{Gini} &= 1 - \sum p_i^2 \\ &] \end{aligned}$$

Information Gain:

$$[$$
$$IG = H(\text{parent}) - H(\text{children})$$
$$]$$

The root feature should be selected based on the **highest Information Gain** or **lowest Gini impurity**.

Example structure:



The exact tree should be obtained from the actual dataset.

(b) Evaluation

The Decision Tree should be evaluated using:

Metric	Value
Accuracy	[Calculate]
Precision	[Calculate]
Recall	[Calculate]
F1-Score	[Calculate]

Formulas:

$$[$$
$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$
$$]$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1} = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

A **False Negative** occurs when a diabetic patient is predicted as non-diabetic.

False Negatives can be reduced using SMOTE, class weights, threshold adjustment, and optimising the model for higher recall.

(c) Post-Pruning

A fully grown Decision Tree can overfit the training data.

Pre-pruning controls tree growth during training using parameters such as maximum depth.

Post-pruning removes unnecessary branches after the tree has been trained.

Cost-complexity pruning uses:

$$R_{\alpha}(T) = R(T) + \alpha |T|$$

Comparison:

Model	Accuracy
Pre-pruned Tree	[Calculate]
Post-pruned Tree	[Calculate]

A **pruned Decision Tree** is generally preferable for clinical use because it is simpler, more interpretable, and less prone to overfitting.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression predicts the probability of diabetes:

$$P(Y=1)=\frac{1}{1+e^{-z}}$$

where:

$$z=\beta_0+\beta_1X_1+\cdots+\beta_nX_n$$

Comparison:

Model	Accuracy	AUC-ROC
Decision Tree	[Calculate]	[Calculate]
Logistic Regression	[Calculate]	[Calculate]

Logistic Regression is preferable when a simple, interpretable model and probability-based risk estimation are required.

(b) Feature Importance

Top three features should be obtained from the trained models.

Rank	Decision Tree	Logistic Regression
1	[Feature]	[Feature]
2	[Feature]	[Feature]
3	[Feature]	[Feature]

If both models identify similar features, those variables likely contain strong predictive information. Differences may occur because Decision Trees can model nonlinear relationships and interactions, while Logistic Regression uses a linear relationship in log-odds.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP

The treatment recommendation problem can be represented as:

[
MDP=(S,A,P,R,\gamma)
]

States

- S0 – Stable
- S1 – Mild Risk
- S2 – Moderate Risk
- S3 – High Risk
- S4 – Controlled

Actions

- Diet
- Exercise
- Medication
- Monitor

Reward

Outcome	Reward
Major improvement	+10
Moderate improvement	+5
No change	0
Deterioration	-5

The patient moves between states depending on the selected action and health outcome.

(b) Q-Learning

Q-Learning updates the Q-value using:

$$[Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]]$$

Assume:

$$[\alpha = 0.5, \quad \gamma = 0.9]$$

Initial Q-table values are zero.

Update 1

For S2 → Exercise:

- Reward = 5
- Maximum next Q-value = 2

$$[Q(S2, Exercise) = 0 + 0.5[5 + 0.9(2)]]$$

$$[\mathbf{= 3.4}]$$

Update 2

For S1 → Diet:

- Reward = 5
- Maximum next Q-value = 3.4

$$[Q(S1, Diet) = 0 + 0.5[5 + 0.9(3.4)]]$$

$$[\\ =\mathbf{4.03} \\]$$

Update 3

For S3 → Medication:

- Reward = 10
- Maximum next Q-value = 4.03

$$[\\ Q(S3, Medication) \\ =0.5[10+0.9(4.03)] \\]$$

$$[\\ =\mathbf{6.8135} \\]$$

Example training episodes:

Episode	State	Action	Reward	Next State
1	S2	Exercise	+5	S1
2	S1	Diet	+5	S0
3	S3	Medication	+10	S2
4	S2	Monitor	0	S2
5	S1	Exercise	+5	S0

Convergence: Training can be considered converged when the maximum change in Q-values becomes smaller than a predefined threshold such as 0.001 for consecutive episodes.

(c) Final Policy

The policy is:

[
 $\pi(s) = \arg\max_a Q(s,a)$
]

Example:

State	Recommended Action
Stable	Monitor
Mild Risk	Diet
Moderate Risk	Exercise
High Risk	Medication*
Controlled	Monitor

*Medication decisions must be reviewed and approved by a qualified healthcare professional.

Ethical Considerations

1. **Patient Safety** – Unsafe recommendations must be prevented.
2. **Bias** – Training data should represent different patient groups fairly.
3. **Explainability** – Healthcare professionals should understand why an action was recommended.
4. **Human Oversight** – Doctors should have final decision-making authority.
5. **Privacy** – Patient data must be securely stored and protected.
6. **Continuous Monitoring** – The model should be regularly evaluated after deployment.

Conclusion

The proposed system combines supervised learning and Reinforcement Learning for diabetes diagnosis and treatment support. Decision Trees provide interpretability, while Logistic Regression provides statistical risk estimation. Q-Learning can learn treatment policies from patient-state transitions.

For clinical deployment, **Recall, patient safety, fairness, explainability, privacy, and human supervision** are more important than Accuracy alone.